

 quarter = 4 equal size pieces.

1 litre = 1.06 ^{cuarto} quarts

4 quarts = 1 gallon

rounded to 10th how many litres per 1 gallon.

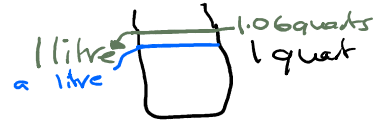
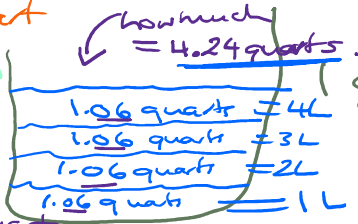
0.943 litre = 1 quart

3.77 litre = gallon

4 quarts = 1 gallon

"a" litres = 1 quart
1 litre = 1.06 quarts

$$a = \frac{1}{1.06} = 0.943$$



a litre = 1 quart
1 litre = 1.06 quarts
a litres = $\frac{1}{1.06}$
= 0.943 litres

in 1 quart
4 quarts = 1 gallon

4 litres = 4 x 0.943 = 3.77 litres (gallon)

To make a drink

6-8 fl oz



6-8 fl oz per pouch mix

3 cups of water to 4 pouches of mix

1 cup = 8 fl oz
1 cup = 8 fl oz
1 cup = 8 fl oz
24 fl oz
4 pouches of mix

how much water do I need for 4 pouches?

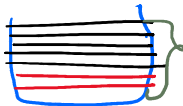
1 cup = 8 fl oz

fl = fluid = liquid
oz = measure = volume.

Unit	Sample Use	Comparing Units
fluid ounce (fl oz)	 1 fl oz	1 fl oz = $\frac{1}{8}$ c
cup (c)	 1 cup 8 fl oz	1 c = 8 fl oz
pint (pt)	 1 pt	1 pt = 2 c
quart (qt)	 1 qt	1 qt = 2 pt
gallon (gal)	 1 gal	1 gal = 4 qt

24 fl oz of water = 6 fl oz
4 pouches of mix = 6 fl oz
pouch of mix

1 gallon = 3.77

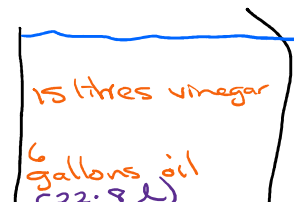


3 x amount

Oil : vinegar
2 gallons = 5 litres
6 gallons = 15 litres
22.8 L = 15 L

3.8 litres = 1 gallon
22.8 L = 6 gallons (oil)

Total litres = 15 + 22.8 = 37.8 litres.



total = 37.8 litres.

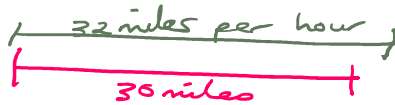
oil Vinegar

2 gallons = 5 litres

3x amount
 $7.54 \text{ litre} = 5 \text{ litres}$
 $22.62 = 15 \text{ litres}$

6 gallons oil
 (22.8 L)

drive 32 mph average place is 30 miles away
 How long to get there?



miles hour
 $32 : 1$
 $30 : x$

hours miles
 $\frac{x}{1} : \frac{30}{32}$

$x = \frac{30}{32}$
 $= \frac{15}{16} = 0.9375$
 Round to 1 hour?

$y \text{ mins} = 0.9375 \text{ hours}$
 $60 \text{ mins} = 1 \text{ hour}$

$60 \times \frac{y}{60} = 0.9375 \times 60$
 $y = 0.9375 \times 60$
 $= 56.25 \text{ mins}$

1. \$150 10% off

$150 = 100\%$
 $15 = 10\%$

$150 - 15 = \$135$



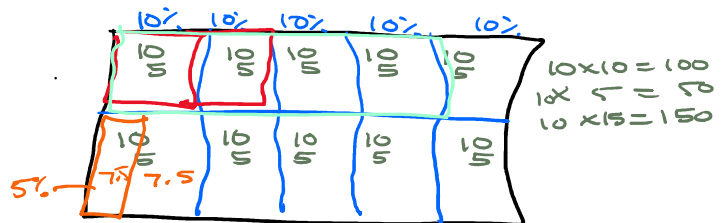
$150 \times 0.9 = 135$

$(150 \times 0.9) = 135$

$150 \times 1 = 150$

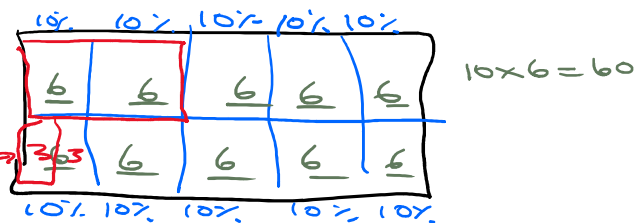
2. What is 10% of \$135

10% of 135 = \$13.50



20% of 150 = 30
 40% of 150 = 60
 5% of 150 = 7.5

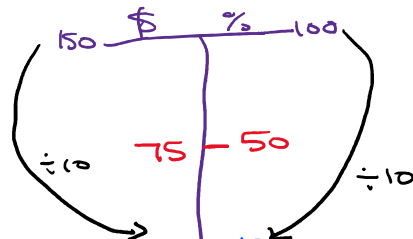
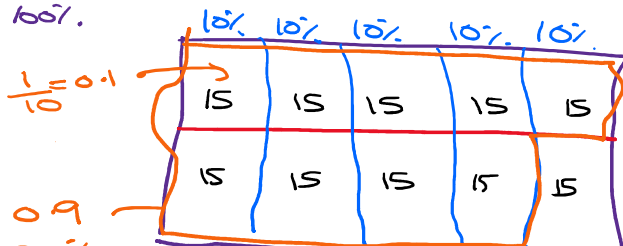
\$60

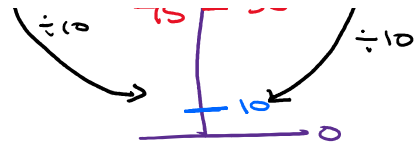
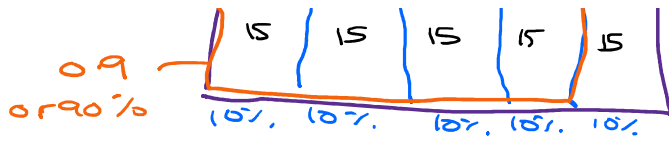


$90\% = \frac{90}{100}$
 $= \frac{9}{10}$
 $= 0.9$

$100\% - 10\% = 90\%$
 $60 - 6 = 54$
 (10% off) = 90%
 $60 = 6$
 $60 = 12$
 $60 = 3$
 $60 = 54$

$150 = 100\%$





90% or $\times 0.9 \Rightarrow 150 \times 0.9 = 135$

$$\begin{array}{r} 2 \overline{) 2x - 6} \\ \underline{2x} \\ -6 \\ \underline{-6} \\ 0 \end{array}$$

$$\begin{aligned} 2(x-3) + 4 &= 3(x+2) - 5 \\ 2x - 6 + 4 &= 3x + 6 - 5 \\ 2x - 2 &= 3x + 1 \\ -2 &= x + 1 \\ -3 &= x \\ x &= -3 \end{aligned}$$

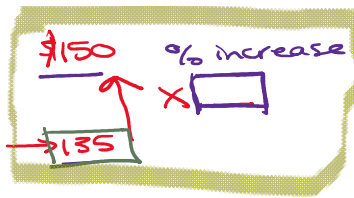
Expand (get rid of the brackets)

$$\begin{aligned} 2x - 2 &= 3x + 1 \\ -2x &= 3x + 3 \\ -5 &= x + 3 \\ -3 &= x \end{aligned}$$

Shop sells \$150, 10% off everything
new price = \$135 (150×0.9 or $150 - 15$)
(90%) 100% - 10%



10% reduction
 $\times 0.9$



10% of 135 is 13.50

$135 + 13.5 = 148.5$
Not \$150 that I started with!!!
 $\frac{15}{150} = \frac{10\%}{100\%} = 10\%$
 $0.9 = 0.90 = 90\%$

\$150 $\rightarrow$ 10% was \$15
so 10% discount = $150 - 15 = 135$

90% of 150 is $150 \times 0.9 = 135$

\$135 what is the % increase to get back to \$150

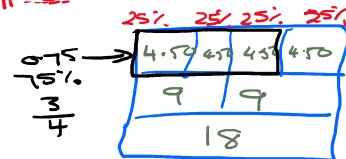
$135 \times \frac{0.112}{100} = 150.12$

1.111 actually mean

$$\begin{aligned} 135 \times \frac{1.111}{100} &= \frac{150}{135} \\ \frac{1.111}{100} &= \frac{150}{135} \\ &= 1.111 \end{aligned}$$

$\frac{1.111}{100} = 1.111\%$

Jacket on Sale! 25% off
original \$18 sale price.



calculate 25% & subtract
 $18 \times \frac{25}{100} = 4.50$
 $18 - 4.50 = 13.50$

multiply to take away 25%

$18 \times 0.75 = 13.50$

$\frac{75}{100} = \frac{3}{4}$
 $75\% = \frac{75}{100} = \frac{3}{4}$

$$18 \times \frac{25}{100} = 4.50$$

$$18 - 4.50 = 13.50$$

Pants on Sale 25% off

original price is \$80

sale price = ?

$$\begin{aligned} 25\% \text{ of } 80 &= \$20 \\ \text{sale price} &= 80 - 20 \\ &= 60 \end{aligned}$$

$$80 \times \frac{75}{100} = 80 \times 0.75 = 60$$

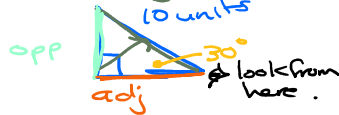
$$10 \wedge [0.15] = 13.50$$

$$75\% = \frac{75}{100} = \frac{3}{4}$$

$$\frac{75}{100} = \frac{3}{4} = 75\%$$

25%	25%	25%	25%
20	20	20	20
40	40		
80			

side opposite the right angle
hypotenuse



opposite = across from
adjacent = next to

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

To find the opposite

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$10 \times \sin(30) = \frac{\text{opp}}{10} \times 10$$

$$10 \sin(30) = \text{opp}$$

$$\text{opp} = 5$$

$$\begin{aligned} \text{opp} &= ? \\ \theta &= 30 \\ \text{hyp} &= 10 \end{aligned}$$

To find the adjacent

$$\begin{aligned} \cos \theta &= \frac{\text{adj}}{\text{hyp}} \\ \cos(30) &= \frac{\text{adj}}{10} \end{aligned}$$

$$10 \times \cos(30) = \frac{\text{adj}}{10} \times 10$$

$$\begin{aligned} \text{adj} &= 10 \times \cos(30) \\ &= 5\sqrt{3} \text{ (Exact)} \\ &= 8.66 \text{ (Approx)} \end{aligned}$$

$$\begin{aligned} \text{adj} &= ? \\ \text{hyp} &= 10 \\ \theta &= 30 \end{aligned}$$

You can make only four different cuboids with 16 cubes.

		Dimensions		
Cuboid A		1	1	16
Cuboid B		1	2	8
Cuboid C		1	4	4
Cuboid D		2	2	4

(a) Which of the cuboids A and D has the larger surface area?

Tick (✓) the correct answer below.

- ☐ Cuboid A
☐ Cuboid D
☐ Both the same

Explain how you know.

(b) Which cuboid has the **largest volume**?
Tick (✓) the correct answer below.

Cuboid A ☐

Cuboid B ☐

Cuboid C ☐

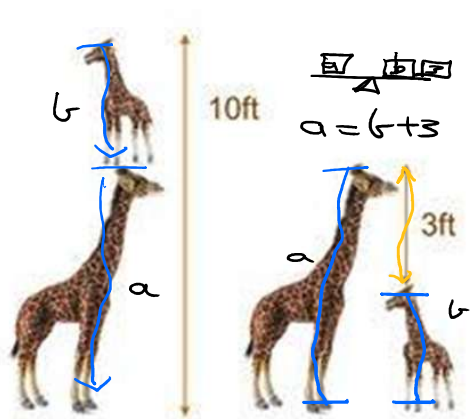
Cuboid D ☐

All the same ☐

(c) How many of cuboid D make a cube of dimensions $4 \times 4 \times 4$?
.....

(d) You can make only six **different** cuboids with 24 cubes.
Complete the table to show the dimensions.
Two have been done for you.

	Dimensions		
Cuboid E	1	1	24
Cuboid F	1	2	12
Cuboid G			
Cuboid H			
Cuboid I			



find height of each giraffe?
 a = height of the adult giraffe
 b = " " " baby giraffe

$$a + b = 10$$

$$a - b = 3$$

$$a + a = 2a$$

$$a \times a = a^2$$



31/1/25

$$a + b = 10$$

$$a - b = 3$$

$$a = 6.5$$

$$6.5 + b = 10$$

$$b = 10 - 6.5$$

$$b = 3.5$$

$$6.5 - b = 3$$

$$6.5 - 3 = b + 3 - 3$$

$$3.5 = b$$

Left LHS $a + b = 10$

Right RHS $2a = 13$

$$a = \frac{13}{2} = 6.5$$

$$\frac{2a}{2} = \frac{13}{2} \quad (\div 2)$$

$$a = 6.5$$

2x + 4y = 25

3x + 3y = 72

6x + 12y = 75

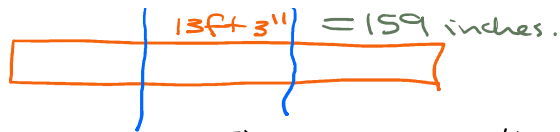
6x + 6y = 144

cat

If a 13-ft board is cut into 3 equal pieces.
How long is each piece?

$$13\text{ft} + 3'' = 159\text{ inches.}$$

Estimate / guess



Estimate / guess

Answer is definitely more than 4 (ft) (3x4=12)
less than 5 (ft) (3x5=15).

13 ft 3 inches $\Rightarrow$ all inches.

$$13 \times 12 \text{ inches} + 3 \text{ inches} = 156 + 3 = 159 \text{ inches}$$

(ft) $\rightarrow$ 1 foot = 12 inches.



$$\begin{array}{r} 53 \\ 3 \overline{) 159} \end{array}$$

so each piece is 53 inches

53 inches = 4 ft + 5 inches.

$$\begin{array}{l} 4 \times 12 + 5 \\ 48 + 5 \end{array}$$

Learning is not event, it is a journey

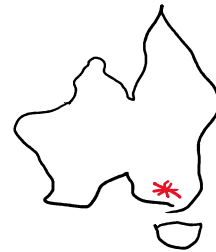
Spain $\Rightarrow$ wonderful maths museum

Stone soup

bean shoot

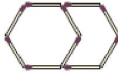
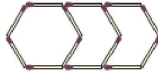
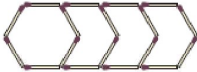
(mung bean) $\rightarrow$ 1 week

oak tree $\rightarrow$ more than your
acorn lifetime



Patterns aka Algebra

Patterns aka Algebra		Number of matches in each pattern						Rule m =	Number of matches in pattern 10		
Draw the next pattern in the sequence		Pattern Number (p)		0	1	2	3	4	5		
 Pattern 1 $3 \times 1 + 1 = 4$  Pattern 2 $3 \times 2 + 1 = 7$  Pattern 3 $3 \times 3 + 1 = 10$  Pattern 4 $3 \times 4 + 1 = 13$ Matchsticks (m) Pattern 5 $\text{matchsticks} = 3 \times \text{pattern} + 1$											
Draw the next pattern in the sequence		Pattern Number (p)		1	2	3	4	5			
 Pattern 1  Pattern 2  Pattern 3  Pattern 4 Matchsticks (m) Pattern 5											
Draw the next pattern in the sequence		Pattern Number (p)		1	2	3	4	5			
 Pattern 1  Pattern 2  Pattern 3  Pattern 4 Matchsticks (m) Pattern 5											

Patterns		Number of matches in each pattern					Rule $m =$	Number of matches in pattern 10
Draw the next pattern in the sequence		1	2	3	4	5		
								
Pattern 1	Pattern 2							
								
Pattern 3	Pattern 4							
Matchsticks (m)								
Pattern 5								
Draw the next pattern in the sequence		1	2	3	4	5		
								
Pattern 1	Pattern 2							
								
Pattern 3	Pattern 4							
Matchsticks (m)								
Pattern 5								

Pattern 1

Pattern 2

Pattern 3

Pattern 4

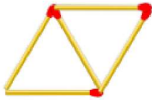
Matchsticks (m)

Pattern 5

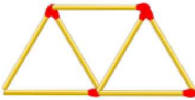
Question 1:



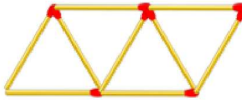
Pattern 1



Pattern 2



Pattern 3



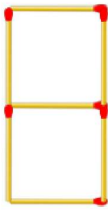
Pattern 4

Pattern Number:	1	2	3	4	...	n	...	100	
Number of Matchsticks:									87

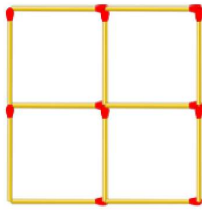
Rule in words: number of matchsticks = _____

Rule in symbols: $m =$ _____

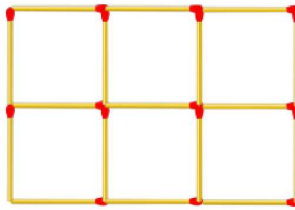
Question 2:



Pattern 1



Pattern 2



Pattern 3

Pattern Number:	1	2	3	4	...	n	...	114	
Number of Matchsticks:									122

Rule in words: number of matchsticks = _____

Rule in symbols: $m =$ _____

Question 3:

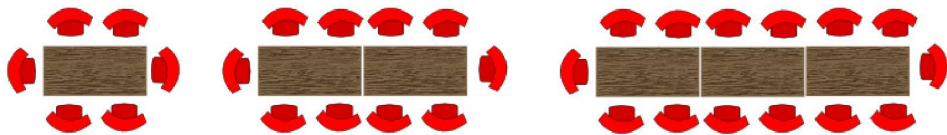


Number of Tables:	1	2	3	4	...	n	...	138
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 4:

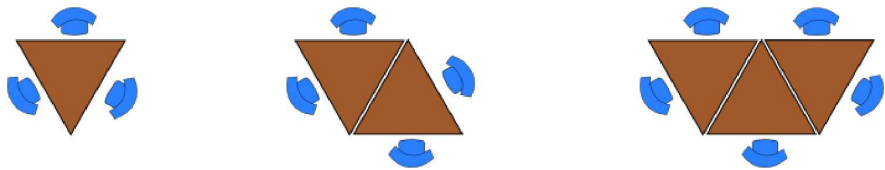


Number of Tables:	1	2	3	4	...	n	...	90
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 5:

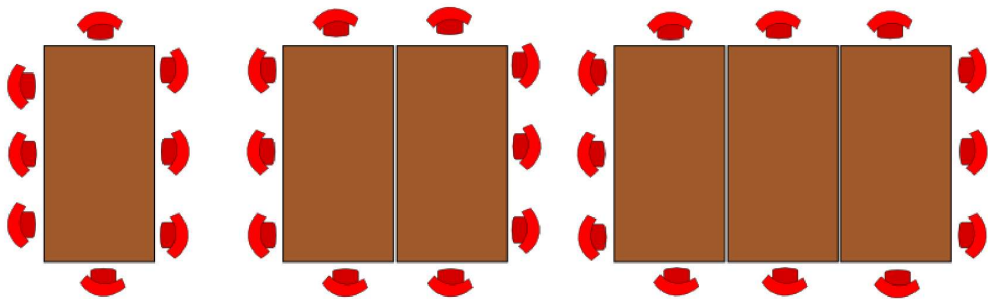


Number of Tables:	1	2	3	4	...	n	...	102
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 6:

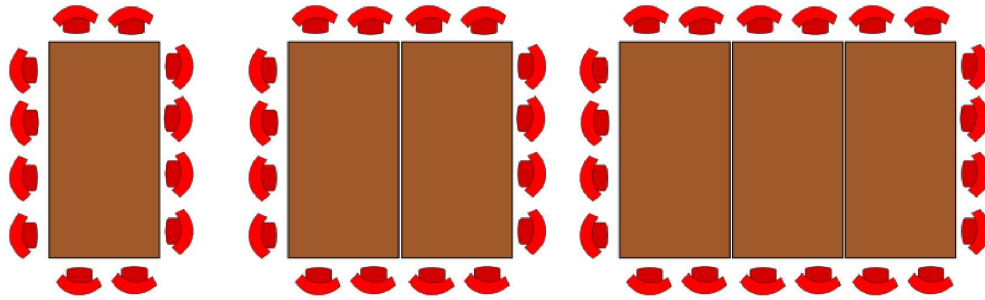


Number of Tables:	1	2	3	4	...	n	...	112
Number of Chairs:								

Rule in words: number of chairs = _____

Rule in symbols: $c =$ _____

Question 7:



Number of Tables:	1	2	3	4	...	n	...	150	
Number of Chairs:									

In the space below, create your own (random!) pattern and ask another student to work out the rule for the pattern: